

ReSolve: Instruction-Guided 3D CAD Editing with Geometry and View Evaluation

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Abstract

001 *Instruction-guided CAD editing must alter the requested*
002 *feature while preserving the rest of a three-dimensional*
003 *shape. Program validity and global volume overlap alone*
004 *cannot establish this behavior. We study executable shape*
005 *editing with ReSolve, a code-conditioned pipeline that pre-*
006 *dicts validated program patches, executes native CAD rep-*
007 *resentations, and evaluates resulting solids and derived*
008 *views. A fresh 3B model is fine-tuned on 2,540 public edit*
009 *pairs from BenchCAD and CAD-Editor. On a locally de-*
010 *finied 252-task holdout, strict volume agreement increases*
011 *from 1 to 39 cases; 139 trained outputs execute success-*
012 *fully, exposing a substantial gap between valid geometry*
013 *and target agreement. A focused audit of previously illus-*
014 *trated outputs compares four common-camera projections*
015 *and source-relative edit masks. A flange example demon-*
016 *strates how high overall silhouette agreement can conceal*
017 *misplaced mounting holes. A separate diagnostic FEM au-*
018 *dit checks four completed prediction/target pairs under ex-*
019 *plicit synthetic material and load contracts; two other at-*
020 *tempts remain unverified. The study contributes an au-*
021 *ditable evaluation workflow and empirical evidence, rather*
022 *than a new CAD architecture. The model receives programs*
023 *and text, with no learned image input or solver supervision.*
024 *Local dataset repartitioning, limited visual coverage and a*
025 *single training seed bound the conclusions.*

026 1. Introduction

027 Changing four mounting holes to six is a shape-editing prob-
028 lem: the hole pattern must change while the flange body,
029 hole diameter and bolt-circle radius remain consistent. Sim-
030 ilar requests remove a bracket flange, add a blind shaft
031 slot or enlarge a bore. These operations require localized
032 changes to three-dimensional geometry, and their effects
033 should remain coherent across views. Figure 1 illustrates
034 two complex released source/target pairs.

035 Executable CAD provides a structured representation for

this task. A model can change a construction history rather
than separately generate each rendered view. Yet executable
code is only one level of success. A valid edited solid may
have the wrong feature locations; a high global overlap score
may hide those errors because the affected region is small.
For visual-geometry research, the central question is how to
evaluate the requested shape change as well as overall shape
agreement.

We implement ReSolve, an instruction-guided program-
editing pipeline with three complementary observations:
patch and execution outcomes, agreement with reference
solids, and source-relative changes in common-camera pro-
jections. A supervised language model reads native CAD
and an instruction, emits an original-coordinate patch, and
executes through a representation-specific backend. Four-
view drawings are derived from the same solid. Visual eval-
uation therefore concerns the executed geometry; it is not an
image-conditioning or drawing-perception experiment.

The primary experiment uses a completed Colab T4 run
with a fresh Qwen2.5-Coder-3B model, trained on public
paired edits from BenchCAD and CAD-Editor [15, 16]. The
resulting 39 strict matches among 252 held-out tasks im-
prove on the fresh base’s one match, but only 39 of 139
executed outputs meet the strict threshold. We retain this
failure gap as a substantive finding. A subsequent visual
audit illustrates why global agreement cannot substitute for
feature-sensitive evaluation.

Mechanical response provides a further, conditional ob-
servation. We apply finite-element checks to illustrative
edits only after specifying units, material, fixtures and
loads. These post-edit checks measure numerical and target-
response consistency under synthetic test scenarios. They
neither provide training feedback nor establish manufactur-
ing safety.

Contributions.

- A documented two-source executable CAD editing exper-
iment, separating patch applicability, solid execution and
reference agreement on every retained held-out task.

- A source-relative multi-view diagnostic over saved edited solids, with a realistic mounting-hole example that exposes the difference between whole-shape similarity and localized edit agreement.

- An auditable link between edited geometry, derived vector views and scenario-defined FEM fields, with explicit unavailable outcomes and analysis assumptions.

Our contribution is empirical evaluation and reproducibility. Standard supervised adaptation, program patches, projections and FEM are not claimed as individually new. Earlier hardware runs, localized transport pilots and deterministic structural controls are documented separately in the supplement.

2. Related work

Structured 3D generation and editing. DeepCAD [14] models parametric construction histories, and Text2CAD [5] generates sequential CAD from text. CAD-Editor [15] directly studies instruction-based sequence editing. BenchCAD [16] releases programmatic mechanical tasks and edit pairs; neuralCAD-Edit [9] evaluates expert-requested multimodal edits with geometry and feature-oriented measures. Our study uses released supervision and a standard code model. It does not reproduce these systems or establish superiority over them. It instead reports what a modest local adaptation achieves and where execution, bulk geometry and projected edit measurements disagree.

Vector views and visual representations. Drawing2CAD [10] connects vector drawings to CAD histories. DeepSVG [1] and StarVector [11] learn vector graphics representations and generation; SVGEEditBench [7] evaluates instruction-based SVG edits. Our projections are deterministically derived from executed CAD. This preserves shared geometric provenance across views, but does not evaluate image understanding, SVG-to-CAD inference or annotation recovery. The added silhouette/change-mask audit is a diagnostic of visual shape changes, not a learned multi-view consistency objective.

Physical evaluation. FEM-Bench [6] evaluates scientific code generation, while CADEngBench [13] studies functional CAD editing and matched linear-static analysis. We use declared diagnostic loads and materials for a small mechanical subset; these properties are not supplied by the public edit pairs. The resulting FEM checks are external post-edit evaluations. No public runnable CADEngBench release was verified in this work, and its results are not presented as our reproduced benchmark.

3. Executable shape editing and evaluation

Figure 2 links a realistic four-to-six mounting-hole request to the actual final 3B patch, edited solid and numerical fields. The source solid is an illustration of the program input, not a model input image.

3.1. Model and patch contract

Given source representation C and instruction u , the model predicts patch p and the executor returns $\hat{C} = \mathcal{E}(C, p)$. Each operation specifies a zero-based original line index, deletion count and replacement-line list. Operations must be typed, ordered, non-overlapping and in bounds. All coordinates address the original source. CadQuery syntax or native command-schema validation follows patch application.

We fine-tune Qwen2.5-Coder-3B-Instruct [4] with NF4 double quantization and completion-only QLoRA [3]. For supervised pairs (C, u, p^*) , only answer tokens contribute to

$$\mathcal{L}_{\text{edit}} = - \sum_{t \in \text{answer}} \log P_{\theta, \phi}(p_t^* | C, u, p_{<t}^*), \quad (1)$$

where θ denotes frozen base weights and ϕ the adapters. No image features, FEM loss, verifier reward or test-based checkpoint choice enters training. The architecture in Fig. 6 documents the native-representation branches.

3.2. Execution and solid agreement

The appropriate CAD backend builds a boundary-representation solid $\hat{B}$. Successful execution requires valid geometry, at least one solid and positive volume. The CadQuery executor permits a restricted set of imports and builtins, rejects private-attribute access and file I/O, and applies a 45-second per-case subprocess budget. This is defense in depth, not a general Python sandbox. Unsupported constructs and operational failures remain separate outcomes.

For target B^* , volumetric intersection-over-union is

$$J_{\text{vol}} = \frac{V(\hat{B} \cap B^*)}{V(\hat{B}) + V(B^*) - V(\hat{B} \cap B^*)}. \quad (2)$$

We use local broad and strict thresholds of 0.95 and 0.99999, respectively. Exact normalized program/sequence and patch matches are complementary diagnostics. None certifies engineering tolerances, preserved constraints or manufacturing suitability.

3.3. Common-camera view and edit measurements

We derive front, top, right and isometric views from each solid. The linked SVG sheets are CAD-derived views, not independently predicted images or production blueprints. For the focused visual audit, tetrahedral FEM is unnecessary: tessellated boundary surfaces are rasterized into binary orthographic silhouettes $S_v(B)$.

Published BenchCAD source / reference edit pairs

twisted bracket | Remove the second flange; retain the twisted web and first-flange holes.

SOURCE

RELEASED TARGET



stepped shaft | Add a centered blind obround slot to the upper shaft face.

SOURCE

RELEASED TARGET



Reference illustrations only. No stress or FEM result is inferred from these shapes. BenchCAD, CC-BY-4.0.

Figure 1. **Published mechanical CAD edit pairs held out in the local protocol.** Top: remove the perpendicular end flange while preserving the twisted web and the first-flange holes. Bottom: add a centered blind obround slot to the upper cylindrical shaft section. These are source/released-target illustrations, not learned model predictions. Both views are exported from solids. Learned predictions are evaluated separately in the primary experiment and diagnostic visual/FEM audits.

For each case and view v , source, target and available predictions share projection axes, image scale and center. Bounds use their union; no individual shape is recentered. We render 512- and 1,024-pixel square images with 0.3 mm tessellation tolerance. This is a diagnostic camera contract, including available output bounds, rather than a predictor-independent camera benchmark.

For source solid B , define reference and predicted changes as

$$\Delta_v^* = S_v(B) \oplus S_v(B^*), \quad \hat{\Delta}_v = S_v(B) \oplus S_v(\hat{B}), \quad (3)$$

where $\oplus$ is binary exclusive-or. We report both silhouette IoU and edit-mask IoU $J_{\text{edit},v} = \text{IoU}(\Delta_v^*, \hat{\Delta}_v)$. Empty unions are undefined and excluded from the edit mean; nonempty unintended changes are retained. This exposes differences in the changed region, but cannot measure occluded/internal features or instruction semantics. Very small change masks are sensitive to rasterization; we disclose their effect rather than infer feature correctness from a scalar average.

3.4. Conditional mechanical response audit

FEM is applied only with an explicit physical contract. The illustrated BenchCAD scenarios assume mm/N/MPa units, $E = 210,000$ MPa, $\nu = 0.3$, a fixed lower 5 % surface band

and an upper-band resultant of 1 N. These are declared test conditions, not recovered properties or operating loads. Native CAD-Editor examples lack established physical scale and are not assigned such results.

First-order tetrahedra solve small-strain 3D isotropic elasticity with $K_e = V_e B_e^T D B_e$ and fixed-degree elimination. We check finite solutions, anchored components, residuals, force/moment balance and $2U = \mathbf{f}^T \mathbf{u}$ to normalized error below 10^{-6} . Refinement targets at most 5 % compliance change; a local response diagnostic compares predicted and target compliance within 10 %. This complements solid agreement and does not certify stress convergence or safety. Full contracts and saved fields appear in Appendix C.

4. Public data and local protocol

Two native representations. The primary split contains 2,540 training, 172 validation and 252 test tasks (Table 1). BenchCAD provides CadQuery source/target programs, natural-language edits and released STEP references [16]. Microsoft’s CAD-Editor provides native six-bit sketch-extrude edit sequences [15]. We retain their different representations and backends rather than treating them as interchangeable histories. CAD-Editor reference solids

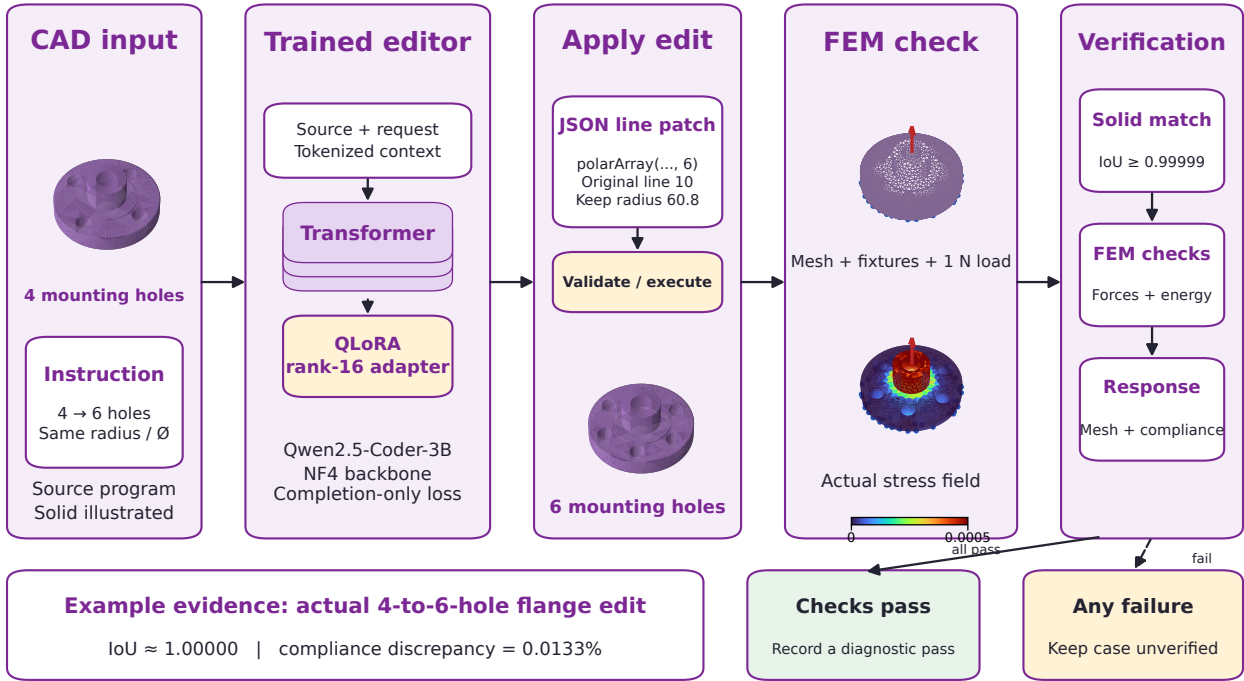


Figure 2. **Instruction-guided executable 3D editing.** Native CAD and text condition the trained editor. Validated patches produce the edited solid; views, solid comparison and optional FEM inspect that result. The shown flange is an actual final 3B prediction changing four holes to six at the same diameter and bolt-circle radius. Its saved tetrahedral mesh, fixture (blue), load (red) and von Mises field use a declared synthetic 1 N test; stress is clipped at p95 in MPa. FEM is an external diagnostic, not training supervision. Failed stages remain unverified.

Source	Train	Validation	Test
BenchCAD	492	44	124
CAD-Editor	2,048	128	128
Total	2,540	172	252

Table 1. **Retained primary data.** Counts define our local protocol, not an official benchmark split. Source-specific reference construction remains distinct.

are reconstructed locally through our decoder; they are not author-released STEP references.

Provenance and isolation. BenchCAD is pinned at revision 5919f578ab09ec283603a082fab07c7639ab56eb and declares CC-BY-4.0. Its 748 edit pairs yield 710 after family-based exclusion of pipe/electrical content. Family groups are joined when source/target AST templates match after numeric-literal masking; whole connected components are assigned to local splits before patch construction. The earlier retained test slice had 126 tasks; two additional pipe-related instructions are excluded from the primary

experiment, yielding 124. CAD-Editor uses a documented subset of the public training/evaluation archive. Exact source/target connected components are isolated within each source; cross-source geometric ancestry is not established.

Retention and supervision. Token filtering and omissions are saved before training and evaluation. The primary model uses a 4,096-token context and a 768-token output cap. Training patches replay their normalized targets exactly. Comments and blank lines are removed when preparing CadQuery patches, avoiding supervision of parameter-comment edits as geometry. Published instructions are retained without adding answer-line hints. Native command sequences use their own schema and decoder.

Evaluation boundaries. BenchCAD’s edit release was originally held out; repartitioning it for training changes the benchmark contract. We report local held-out agreement, not official leaderboard or official-test generalization. The primary denominator includes all 252 retained tasks, including six reference errors and one final-model worker timeout. Reference availability is tracked separately. A previously

identified instruction/target conflict in a shaft-slot record illustrates that reference agreement does not certify natural-language compliance for every label. The supplement preserves the earlier split details and audits.

5. Experiments

5.1. Primary two-source adaptation

A fresh 3B backbone uses rank-16 adapters with alpha 320 on seven attention/MLP projections in the last twelve transformer blocks. Seed 17, learning rate 10^{-4} , microbatch one and four accumulation steps give effective batch four. Two fixed epochs on a Colab T4 complete 1,270 optimizer updates in 20,923.7 seconds. Mean completion loss is 0.2495. The saved manifest records `torch.bfloat16`; native BF16 hardware acceleration is not established. We reload the final adapter, check all 168 saved tensors for finite values and evaluate both fresh base and trained model greedily on identical held-out inputs. The 9,977,856 adapter scalars and artifact hashes are saved.

Executable is not necessarily correct. Table 2 separates interface success from shape agreement. Fine-tuning increases applicability from 39 to 173 tasks and execution from 7 to 139. Strict agreement rises from 1/252 to 39/252 (15.48 %), with 25/124 BenchCAD and 14/128 CAD-Editor matches. Only 39/139 executed outputs (28.06 %) reach the strict threshold, leaving considerable geometric error. Broad agreement reaches 52/252. Exact target program/sequence and patch agreement are 32 and 31 cases, respectively, versus one each for the base. No final generation reaches the output cap.

Six reference errors and one whole-worker timeout remain in the full denominator. The scorer’s reference-valid bookkeeping differs between base and final outputs (246 versus 245); 245 is therefore not treated as a fixed valid-reference test set. These counts measure released-target agreement, not audited compliance with every instruction.

What this comparison establishes. The same-model base/final comparison supports a local adaptation gain under fixed prompts and decoding. It is not a matched comparison with CAD-Editor or other published editing methods. One seed, synthetic online supervision, token filtering and unestablished cross-source ancestry limit generalization. Earlier 1.5B Apple and H100 runs use different data/settings and are preserved as separate supplementary experiments (Appendix G); their counts are not pooled with the primary run.

5.2. Frontier comparison, plan anchoring and verifier selection

Frontier model on the same holdout. We evaluate GPT-6 Astra [8] zero-shot on all 252 held-out tasks with the identical system prompt, input, patch contract and scorer (medium reasoning effort, one sample per task, \$5.15 at list price). It reaches 106/252 strict matches (89 BenchCAD, 17 CAD-Editor), against 39 for the final 3B adapter. On paired tasks, only Astra succeeds on 70 and only the adapter on 3 (exact McNemar $p = 1 \times 10^{-17}$). The gap lies in BenchCAD programs: on CAD-Editor sequences the two are statistically indistinguishable (6 versus 3 discordant tasks, $p = 0.51$), where 77 frontier answers execute but disagree with the reconstructed target and 31 violate the patch or six-bit command contract. Some frontier misses follow the instruction more literally than the released target, e.g. a requested 21 mm through-hole against a released 20.98 mm hole; strict agreement measures target reproduction, not instruction compliance. One execution-feedback repair round, in which Astra sees its failed answer and the validator or executor error but no reference, makes 16 of its 32 failed patches valid and 14 executable, yet adds only 1 strict match (low reasoning effort, \$0.64).

GRPO-trained verifier for Astra. We also train a separate verifier policy with GRPO [12]: a Qwen2.5-Coder-1.5B LoRA policy chooses among four constrained actions (execute the candidate, check instruction numbers, accept, reject) and must observe at least one tool result before a verdict. It learns from 726 saved Astra candidates on BenchCAD training tasks, with 238 component-disjoint development candidates, over 600 groups of 8 trajectories; reference agreement defines the reward but never enters its observations. Each held-out Astra answer it rejects receives one repair request, and a repair replaces the original only if the policy accepts it. On the 124 BenchCAD held-out tasks, this raises Astra from 89 to 91 strict matches (2 recovered, 0 regressed, $p = 0.50$; 60 repair calls, \$0.79). The verifier itself is weak: its held-out verdict accuracy (57.7 %) is below always accepting Astra’s answer (72.4 %), and on development it matches that baseline (81.9 versus 81.5 %). The small gain is therefore better attributed to the extra repair calls than to learned verification.

Plan-then-patch with quote anchoring. We retrain the same 3B recipe with each answer formatted as a short plan followed by the unchanged patch. One plan bullet per operation states the zero-based original index, quotes that original line and names the action. Template plans are generated deterministically from reference patches, so this format adds no supervision beyond the patch. Decoded verbatim, the plan model is no better than the patch-only adapter (36 and

Model / source	Tasks	Applicable	Executed	IoU ≥ 0.95	IoU ≥ 0.99999
Fresh 3B base, all	252	39	7	2	1
Final 3B adapter, all	252	173	139	52	39
Final, BenchCAD	124	91	65	32	25
Final, CAD-Editor	128	82	74	20	14

Table 2. **Primary held-out outcomes.** Counts retain all 252 tasks, including unavailable reference comparisons and the final-model worker timeout. Strict is a local volume threshold. BenchCAD uses released STEP targets; CAD-Editor uses locally reconstructed command targets.

System	GPU	Test strict (of 252)				Val. (172)
		Dec.	Anch.	Bench	Editor	
Fresh 3B base	T4	1	–	1	0	–
3B adapter, patch	T4	39	–	25	14	–
GPT-6 Astra	API	106	–	89	17	–
+ one repair round	API	107	–	89	18	–
Plan, template (run 1)	L4	36	68	51	17	–
Plan, template (run 2)	A100	42	68	49	19	40
Plan, best of 11 (verifier) [‡]	H100	–	81	55	26	49

Table 3. **Frontier comparison and plan-anchored 3B students** on identical held-out tasks (strict volume IoU ≥ 0.99999). *Dec.*: decoded answers; *Anch.*: anchored, which relocates each patch operation to the unique source line quoted in its plan. Bench/Editor split the rightmost test count (BenchCAD/CAD-Editor). Validation counts are anchored and scored against executed reference programs. Plan rows use the 3B recipe of Sec. 5, one training run each; run 2 differs from run 1 in 10 of 2,540 training targets. [‡]Verifier choice among the greedy and 10 sampled anchored answers of one plan student.

42 strict in two training runs). The first run’s 31 out-of-range BenchCAD patches share one failure: the plan quotes the correct source line but miscounts its index, *e.g.* the intended cut line labelled 21 in a 19-line program. Quote anchoring therefore moves each operation to the unique source line equal to its quoted text and otherwise keeps the predicted index; it reads only the source program and the model output. Anchoring raises both runs to 68/252 (32 and 26 tasks gained, none lost), against 39 for the patch-only adapter (first run: 35 gained, 6 lost, $p = 5 \times 10^{-6}$). Because the rule was designed from first-run test errors, we confirm it on the 172 validation tasks of the second run, which played no role in its design: strict agreement rises from 30 to 40 (10 gained, 0 lost, $p = 2 \times 10^{-3}$). The two runs disagree on 8+8 tasks with equal totals, so single-run differences of a few tasks are not meaningful.

Verifier-selected sampling. Unlike Astra, the plan student produces diverse answers. Besides its greedy answer we sample 10 answers per task at temperature 1.0 and anchor each, giving 7.9 distinct programs per test task on average. A target-free verifier selects one candidate from the source program, instruction and candidates alone: a can-

didate must apply and execute to a valid solid, values requested in the instruction should appear in its inserted lines, and candidates are clustered by executed geometry (volume IoU ≥ 0.99999), preferring the largest agreeing cluster. If no candidate executes, the greedy answer is kept. Selection raises strict agreement from 68 to 81/252 on test (14 gained, 1 lost, $p = 1 \times 10^{-3}$) and from 39 to 49 on validation (10 gained, 0 lost, $p = 2 \times 10^{-3}$); majority vote reaches 72, and always choosing a correct candidate when one exists would reach 90. On CAD-Editor sequences the selected student exceeds Astra (26 versus 17; 10 versus 1 discordant tasks, $p = 0.01$). The same selection barely helps Astra: on 462 BenchCAD training tasks with two Astra samples each, choosing between them changes strict agreement from 368 to 370 (at most 374), because its samples rarely differ. The sampled student is a plan model whose 441 BenchCAD training targets are verifier-selected Astra answers; this substitution leaves its greedy agreement equal to template training (68 versus 68).

Execution-feedback repair for the student. We also let the plan student repair its own failures in a ReAct-style loop. An answer that fails to apply or execute is returned with the task, the failed answer and the validator or executor error, and the student regenerates greedily, for up to 3 rounds. Across the 424 test and validation tasks, answers passing both checks rise from 300 to 313 after one round and do not change afterwards. Strict agreement moves from 68 to 69 on test (1 gained, 0 lost) and stays at 39 on validation. Without repair-specific training, the student rarely turns an error message into a correct edit; sampling with verifier selection is far more effective.

Difficulty tiers. We group held-out tasks by the reasoning an instruction requires, independently of model outputs (Table 4); per-category results were known when the grouping was fixed, so it is descriptive rather than preregistered. On easy edits the selected student is statistically indistinguishable from Astra (64 versus 59, $p = 0.33$). The gap lies in moderate edits that need derived values or feature changes (12 versus 30, $p = 3 \times 10^{-4}$), while hard edits remain largely unsolved by every system (at most 17/118). Shape complexity does not order difficulty: on equal-size

Tier (tasks)	Patch	Plan	Best of 11	Astra
Easy (84)	35 (42%)	55 (65%)	64 (76%)	59 (70%)
Moderate (50)	1 (2%)	8 (16%)	12 (24%)	30 (60%)
Hard (118)	3 (3%)	5 (4%)	5 (4%)	17 (14%)

Table 4. **Held-out strict agreement by edit tier.** Tiers follow what the instruction requires: easy = stated values or removal (BenchCAD T1/T2, CAD-Editor deletions); moderate = derived values or standard feature changes (T3/T4); hard = new geometry from a description (T5, CAD-Editor additions and modifications). Patch: patch-only 3B adapter; Plan: anchored plan student; Best of 11: verifier selection (Table 3).

tertiles of source-solid face count, Astra reaches 35, 50 and 43 %, and the selected student 23, 39 and 37 %, from fewest to most faces.

5.3. Focused multi-view edit audit

The post-hoc audit uses the fixed six previously illustrated held-out cases, selected before this visual measurement. All six older H100 solids are available; only four completed final 3B outputs are eligible. The 3B propeller patch fails syntax and the screw pipeline is incomplete. We do not assign them projected geometry scores or treat four available cases as six successes.

Figure 3 examines a functional flange edit. For the older prediction, top-view silhouette IoU is 0.908 but source-relative edit-mask IoU is 0.516. The final 3B output is 1.000 for both at three-decimal precision. Averaging whole silhouettes across views would further obscure a localized hole-pattern error. This supports examining the requested changed region alongside bulk volume and overall projected similarity; it does not establish general feature detection accuracy.

Four-view records, STEP hashes and 512/1,024-pixel sensitivity are provided in Appendix E. One-pixel reference-change masks in the flange isometric view illustrate an important limitation: a nearly perfect solid can receive a poor edit-mask score in a view with effectively no visible change. We therefore report the top view and raw denominators, and treat aggregate projected edit scores as diagnostic rather than a certification.

5.4. Mechanical examples with response checks

We attempt the same six illustrative instructions with unmodified final 3B patches. Mounting angle, locator block, shaft and flange complete strict geometry, numerical, mesh-volume and target-compliance checks. Their compliance discrepancies are 0.001123 %, 0.046691 %, 0.009141 % and 0.013270 %, respectively. The invalid propeller and incomplete screw attempts remain unverified; the screw has no saved original FEM verdict establishing its cause. Figure 7 shows saved meshes, displacement and stress fields

for two completed functional edits. This subset establishes workflow feasibility under diagnostic contracts, not full-benchmark FEM coverage or real operating safety.

6. Limitations and conclusion

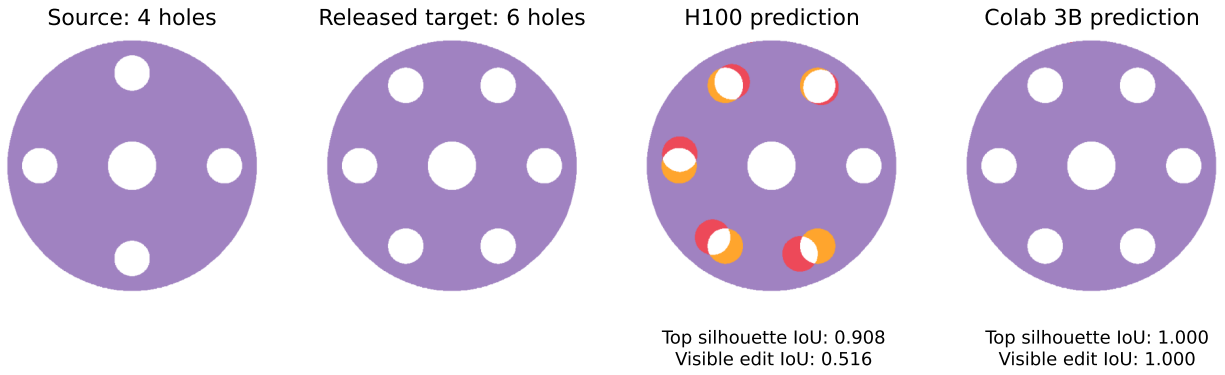
ReSolve studies instruction-guided executable 3D editing through program, solid, projected-change and conditional response measurements. The completed 3B adaptation improves local strict agreement to 39/252, while its large execution-to-agreement gap identifies remaining edit failures. The flange audit shows how localized visual changes can be missed by whole-shape similarity. Saved geometry and FEM fields make successful illustrative outcomes inspectable.

The model is code-conditioned, and deterministic projections do not establish learned visual perception. The visual audit covers six selected cases, with only four available final 3B outputs; it is not a benchmark-wide metric study. Projected masks miss hidden geometry, depend on rasterization and do not certify feature tolerances. Source-specific target reconstruction, known label defects, local repartitioning and unknown cross-source ancestry further limit the protocol. A single seed and standard QLoRA workflow cannot substantiate a new editing architecture or superiority to published CAD methods.

FEM checks use synthetic contracts on a small subset; they do not establish physical properties, stress convergence or operating safety. Stronger conclusions require matched CAD-editing baselines, repeated training runs, full-test feature/view evaluation and physical tasks with independent analysis contracts. Genuine drawing-conditioned editing remains future work rather than a capability of the current model.

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Common top-view camera. Red: target-only silhouette; orange: prediction-only silhouette. Actual saved solids.

Figure 3. **Localized feature agreement versus whole-shape similarity.** The request changes four mounting holes to six while keeping hole diameter and bolt-circle radius. All panels use actual saved solids and the same top-view camera. The older H100 prediction misplaces holes despite relatively high silhouette agreement; source-relative change masks expose the mismatch. The final 3B prediction agrees with the released target in this view. Red/orange show target-only/prediction-only silhouette pixels. This is one illustrative task, not a full-test visual accuracy or a controlled cross-model comparison.

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A. Reproduction and evidence contracts

Online source. The `edit-bench` parquet is downloaded from <https://huggingface.co/datasets/BenchCAD/BenchCAD> at the revision stated in Sec. 4. The original published split name is `edit_bench`. Its 748 rows are not silently treated as 748 independent source shapes; grouping uses all original and edited programs before split assignment. Numeric literals are masked for template linkage; all records of a linked component retain one split.

```
{"edits": [
  {"start": 5, "delete": 1,
   "insert": ["    .hole(6.0)"]}
]}
```

The zero-based line coordinate addresses the original code. A replacement list is required even for one line. Insertion uses zero deletions. The executor rejects overlapping operations and embedded newline characters. This example describes the interface, not an online gold answer.

Saved artifacts. The online dataset manifest pins the parquet checksum, split checksums, source IDs, categories and reference STEP hashes. The run manifest pins model revision, package versions, seed, optimization command, retained rows and omissions. The baseline and reloaded final adapter use identical test prompts. Predictions remain unmodified; the bare-list transport control is saved as a separate diagnostic. Geometry scores include every retained case, including invalid patches, code errors, unsupported constructs, timeouts and Boolean-comparison errors.

Evidence separation. The 150-pair localized pilot, 710-pair online protocol and historical 40,000-edit structural audit are different datasets and contracts. Their results are never pooled. The localized pilot's family test contains only 12 context-retained examples and gives line/number hints. The earlier single-source experiment evaluates 126 context-retained cases; the primary two-source 3B run evaluates 252 tasks. Structural FEM is only established for specified pin-jointed trusses.

Validation. Regression checks test multi-operation original-coordinate execution, insertion/deletion, malformed indexing, overlap rejection and comment preservation. Dataset checks require train/validation/test family and template-component disjointness, exact target-patch replay and pinned split hashes. Final adapters are reloaded for generation and checked for finite tensors. Reference STEP readback and all output geometry scores are retained.

Source/reference illustrations are labeled as such and do not depict model predictions.

B. Held-out model edit examples

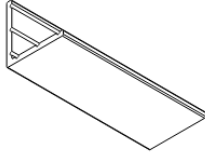
Figures 4 and 5 show six actual predictions from the final three-epoch H100 adapter, alongside the input solid and released STEP target. These are illustrative cases selected after evaluation, not a random sample or additional test set; all 126 evaluated cases remain included in the reported aggregate results. We select three strict matches and three executable but geometrically incorrect edits to expose both supported behavior and remaining limitations. Each view is independently fitted to its panel, so apparent drawing size is not a dimensional comparison. Instructions and measured volume IoU accompany every row. These shape plates alone provide no FEM evidence; the separate scenario-defined audit below supplies the material, loading, support and solver records.

Held-out H100 CAD edits: strict matches

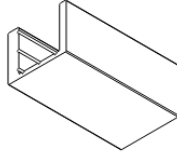
mounting angle box x f130 | T1 | IoU 1.00000

Increase the box X dimension from 33.0 to 42.9.

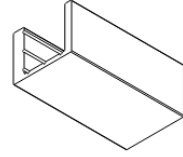
SOURCE



RELEASED TARGET



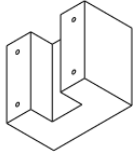
H100 PREDICTION



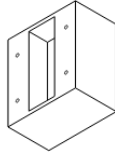
locator block box y f130 | T1 | IoU 1.00000

Increase the box Y dimension from 82.2 to 106.86.

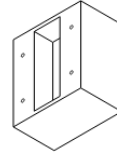
SOURCE



RELEASED TARGET



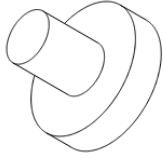
H100 PREDICTION



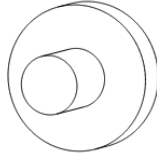
t3medplus stepped shaft rot diag60 | T2 | IoU 1.00000

Rotate the entire part by 60 degrees about the axis (1, 0, 1).

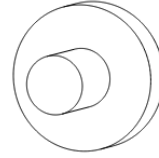
SOURCE



RELEASED TARGET



H100 PREDICTION



Views fit independently; use measured IoU, not apparent drawing size. BenchCAD CC-BY-4.0. FEM audit reported separately.

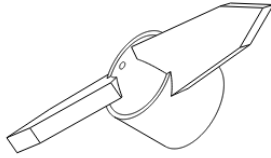
Figure 4. **Three successful held-out H100 edits.** Rows show a mounting-angle dimension change, locator-block width change and a 60-degree stepped-shaft rotation. Columns contain the source, author’s released target and actual final-adapter prediction. Each meets strict volume $\text{IoU} \geq 0.99999$. These examples demonstrate reference agreement within the disclosed code-conditioned protocol; they do not certify general engineering correctness.

Held-out H100 CAD edits: partial / failed edits

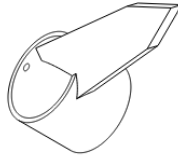
propeller bore widen compute | T4 | IoU 0.63155

Remove one of the two propeller blades; keep the hub, spinner, and the remaining blade.

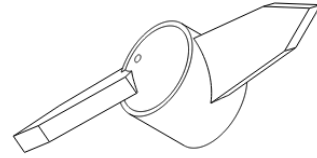
SOURCE



RELEASED TARGET



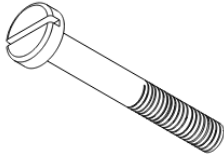
H100 PREDICTION



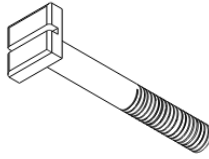
t5hard pan head screw U a | T5 | IoU 0.58449

Replace the round pan head of the screw with a square head of the same across-flats width and the same thickness, keeping the threaded shaft, the screwdriver slot, and the rounded top edge unchanged.

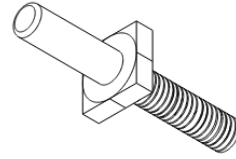
SOURCE



RELEASED TARGET



H100 PREDICTION



t5hard round flange V a | T5 | IoU 0.92018

Change the ring of 4 polar mounting through-passages on the flange face to a ring of 6 mounting through-passages of the same diameter, evenly spaced around the central axis at the same radius.

SOURCE



RELEASED TARGET



H100 PREDICTION



Views fit independently; use measured IoU, not apparent drawing size. BenchCAD CC-BY-4.0. FEM audit reported separately.

Figure 5. **Three executable but incorrect held-out H100 edits.** Removing a propeller blade, replacing a screw head and changing a flange mounting-hole pattern yield valid solids but fail the strict reference threshold. Their volume IoUs are approximately 0.632, 0.584 and 0.920. Source, released target and model output remain separate; no reference operations repair these predictions. Small feature errors can remain important even when bulk volume overlap is high.

C. Scenario-defined FEM audit of mechanical edits

Figure 7 gives a visual walkthrough using two actual final 3B edited geometries; its mesh, displacement and stress panels are linked to saved solver fields.

Scope and physical contract. We attempt FEM on every source, released target and actual prediction in the six illustrated H100 cases: 18 configurations, independently of model training. The released dataset supplies no operating load case. We explicitly assume millimetres, newtons and MPa, a homogeneous isotropic material with $E = 210,000$ MPa and $\nu = 0.3$, and a 1 N resultant tensile test load. These are synthetic diagnostic conditions, not recovered material properties or certified operating conditions. All displacements are fixed at exterior mesh nodes in the lower 5% of the shape’s projected span along the analysis axis. A uniform vector traction is integrated over exterior triangles whose centroids occupy the upper 5%; its area-weighted nodal forces sum to 1 N. The same geometric fixture rule is applied separately to each shape, rather than claiming identical world-coordinate contact patches. The shaft’s fixture and load axis co-rotate with its requested 60-degree pose edit. Every case contract and STEP hash is saved before observing FEM responses.

Discretization and numerical checks. Gmsh [2] meshes the actual STEP geometry with first-order tetrahedra. A SciPy sparse implementation solves small-strain 3D isotropic elasticity,

$$\epsilon = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T), \quad \sigma = \lambda \text{tr}(\epsilon) \mathbf{I} + 2\mu \epsilon. \quad (4)$$

The element stiffness is $K_e = V_e B_e^T D B_e$, with engineering shear strains. Dirichlet elimination solves the free-degree system. Checks require finite displacements, anchored connected components, positive strain energy, free residual, force and moment balance, and the identity $2U = \mathbf{f}^T \mathbf{u}$, each normalized numerical error below 10^{-6} . An analytical uniform-tension cube agrees to numerical precision; load/modulus scaling and co-rotated-fixture tests also pass.

Mesh sensitivity and response comparison. We report compliance $C = \mathbf{f}^T \mathbf{u}$ in Nmm. Global mesh sizes start at one twelfth and one eighteenth of the case’s maximum bounding-box extent, with further refinement if the relative compliance change exceeds 5%. The propeller source requires a third mesh. A local target-response diagnostic accepts compliance discrepancy at most 10%, but this does not replace strict solid agreement. Stress fields and volume-weighted 95th-percentile von Mises values are retained for inspection; compliance convergence does not certify stress convergence. Clamp singularities, small feature

errors, yield, buckling, fatigue, contact and actual-use safety are outside this experiment.

Five of six case triples yield 15 solved source/target/prediction configurations with compliance mesh changes below 5%. Across their recorded meshes the largest normalized equilibrium/energy error is $8.20\text{e-}13$; final CAD-to-mesh volume error is at most 1.39%. Three predictions meet both strict geometry agreement and the local 10% target-compliance agreement tolerance. The propeller and flange predictions fail geometry while meeting that response tolerance. The screw has no FEM response because its boundary mesh fails; this also affects the source and released target, so it is not attributed solely to the model. The incorrect screw output contains seven CAD solids, versus one in its target. Boundary meshing encounters intersecting facets/segments; a source-only finer-mesh and alternative HXT-mesher diagnostic also fails. No component fusion, thread removal, contact model or geometry repair is used to manufacture a solution. This case remains *not verified*.

D. Multi-source Colab training pipeline

The reproducible extension uses paired edits from Bench-CAD and Microsoft’s CAD-Editor in their native representations. The prepared split contains 2,540 training, 172 validation, and 252 test records: respectively 492/44/124 Cad-Query pairs and 2,048/128/128 native sketch-extrude pairs. Two additional pipe-related held-out instructions are excluded from this extension; the earlier 126-case experiment is unchanged. Exact source/target connected components are isolated within each source. Cross-source geometric ancestry is not established. Token-length filtering is recorded before training and evaluation.

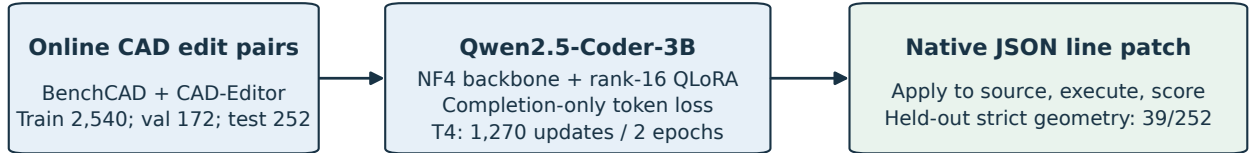
A fresh Qwen2.5-Coder-3B-Instruct backbone uses NF4 double quantization and completion-only QLoRA: rank 16, alpha 320, the seven attention/MLP projections in the last twelve blocks, two fixed epochs, seed 17, learning rate 10^{-4} , effective batch four, a 4,096-token context, and a 768-token output cap. The completed Colab T4 run uses microbatch one with four accumulation steps. Its saved manifest records `torch.bfloat16`; native BF16 hardware acceleration is not established. The final adapter is reloaded for greedy evaluation. Training completes 1,270 optimizer steps in 20,923.7 seconds with mean completion loss 0.2495. The adapter contains 9,977,856 trainable scalars; all 168 saved tensors are finite and their checksum matches the run manifest.

The model maps an instruction and original native CAD to an original-coordinate JSON line patch. Representation-specific schema and syntax checks precede execution. Cad-Query patches are compared to released STEP targets; CAD-Editor command patches execute through a six-bit sketch-extrude decoder, with locally reconstructed target solids explicitly distinguished from author-released STEP

Example	Source C	Target C	Prediction C	Discrepancy (%)	Combined diagnostic
Mounting angle	2.043e-06	7.484e-07	7.464e-07	0.259	Geometry + response agree
Locator block	5.185e-08	3.742e-08	3.742e-08	0.003	Geometry + response agree
Rotated shaft	5.287e-07	5.287e-07	5.291e-07	0.068	Geometry + response agree
Propeller	3.334e-07	5.242e-07	5.221e-07	0.409	Geometry fails; response agrees
Pan-head screw	—	—	—	—	Not verified: mesh failure
Round flange	1.208e-07	1.212e-07	1.212e-07	0.021	Geometry fails; response agrees

Table 5. **Mechanical example FEM checks under explicit test scenarios.** Compliance values use N mm. Discrepancy compares the prediction with the target, with independently generated meshes. A response match is not sufficient for a correct edit. Dashes denote failed meshing, not zero displacement or an invented solver pass.

CAD patch model: multi-source supervised training



FEM is an external post-edit check; its solver outputs do not supervise token training.

Figure 6. **CAD editor architecture and supervised training.** Paired native CAD edits train the quantized Qwen2.5-Coder-3B backbone with completion-only QLoRA. Patches are applied and independently scored on held-out geometry; FEM remains an external post-edit check rather than a training loss.

Model / source	n	Patch	Solid	Strict
Fresh 3B base, all	252	39	7	1
Final 3B adapter, all	252	173	139	39
Final, BenchCAD	124	91	65	25
Final, CAD-Editor	128	82	74	14

Table 6. **Completed two-source Colab evaluation.** Patch denotes applicable schema/syntax; solid denotes successful execution/comparison; strict denotes volume IoU ≥ 0.99999 . All retained tasks remain in n , including six reference errors and one final-model worker timeout. BenchCAD references are released STEP files; CAD-Editor targets are locally reconstructed native sequences.

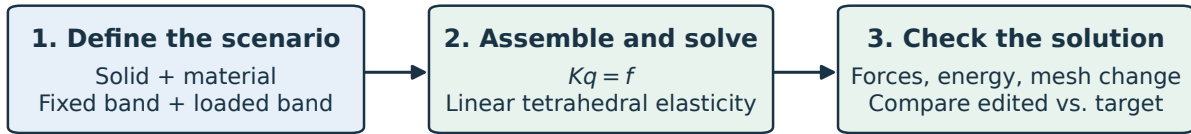
Completed held-out results. The final adapter obtains 39/252 strict matches (15.48%), versus 1/252 for the fresh base. Counts by source are 25/124 for BenchCAD and 14/128 for CAD-Editor. At IoU ≥ 0.95 , the base and final adapter match 2 and 52 cases. Exact target-program/sequence agreement is 1 versus 32; exact patch agreement is 1 versus 31. No final generation reaches the output cap. The whole-worker timeout makes the scorer’s reference-valid bookkeeping model-dependent (246 base, 245 final); we therefore use the complete held-out denominator, not 245 as a reference-valid test set. Different datasets, model size and training settings prevent treating this run as an isolated improvement over the earlier 1.5B experiment.

files. Geometry and failure counts remain separate by source. Only examples with explicit material, unit, fixture, and load contracts proceed to tetrahedral FEM. Equilibrium, energy identity, mesh-volume fidelity, and compliance sensitivity checks precede a combined geometry/response diagnostic. FEM is an external post-edit tool, not a language-generated pass label or a solver reward used in training. Invalid patches and failed meshes remain unverified. The notebook saves adapters, manifests, predictions, geometry metrics, and numerical fields before downloading an artifact bundle.

FEM of the new Colab adapter. We attempt the same six illustrative held-out instructions with the new model’s unmodified patches. Four prediction/target pairs—mounting angle, locator block, rotated shaft, and round flange—pass strict geometry and the scenario-defined numerical, mesh-volume and compliance checks. Their relative compliance discrepancies are respectively 0.001123 %, 0.046691 %, 0.009141 % and 0.013270 %. The flange is now a strict match; the older H100 flange in Fig. 10 remains an incorrect prediction from a different adapter. The new propeller patch

730 fails syntax validation. The screw attempt leaves an incom-
731 plete pipeline directory; its resumed audit records failure,
732 with no saved FEM solution establishing an original cause.
733 Both cases remain unverified. Saved meshes and displace-
734 ment/stress fields substantiate the four passes; source fields
735 are from the separate earlier audit, not recomputed for this
736 pairwise Colab audit. These are diagnostic test scenarios,
737 not operating-safety evidence or full-benchmark FEM cov-
738 erage.

FEM: from an edited CAD solid to a checked response



Blue: fixed nodes Red arrow: resultant load direction All surfaces are undeformed.

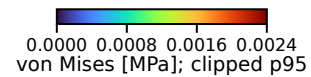
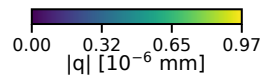
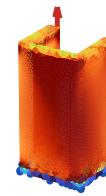
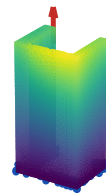
Mounting-angle width edit | actual final 3B prediction

33,871 nodes; 142,334 tets | $C = 7.461\text{e-}07 \text{ N mm}$ | $\max |q| = 9.750\text{e-}07 \text{ mm}$

Tetrahedral surface mesh

Displacement magnitude

Diagnostic stress field



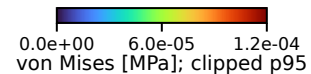
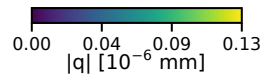
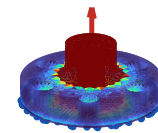
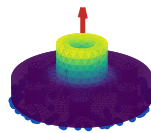
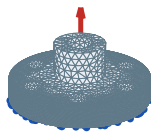
Flange: 4-to-6 mounting-hole edit | actual final 3B prediction

15,035 nodes; 59,326 tets | $C = 1.214\text{e-}07 \text{ N mm}$ | $\max |q| = 1.331\text{e-}07 \text{ mm}$

Tetrahedral surface mesh

Displacement magnitude

Diagnostic stress field



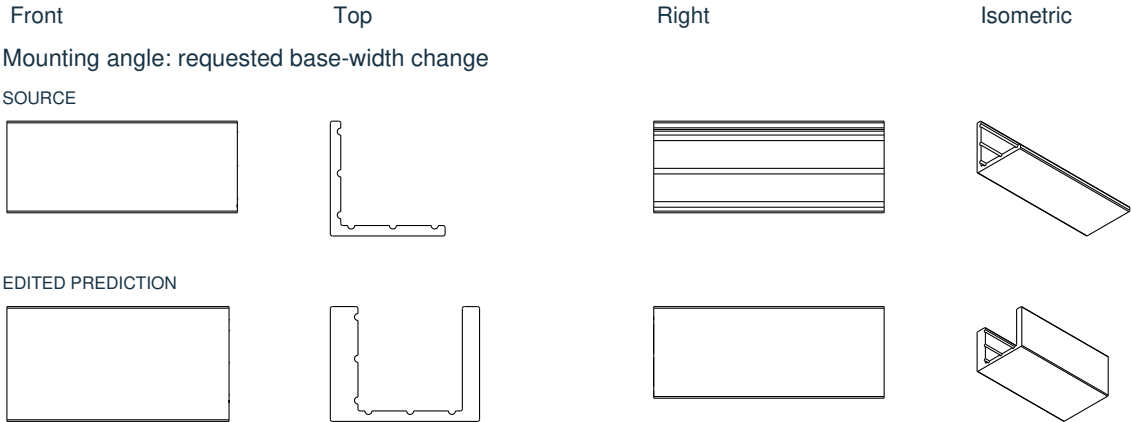
Worked-example interpretation

The same solid supplies the mesh and both field plots. A color change is a solver response, not proof that the edit is correct: independent solid and target-response checks are still required.

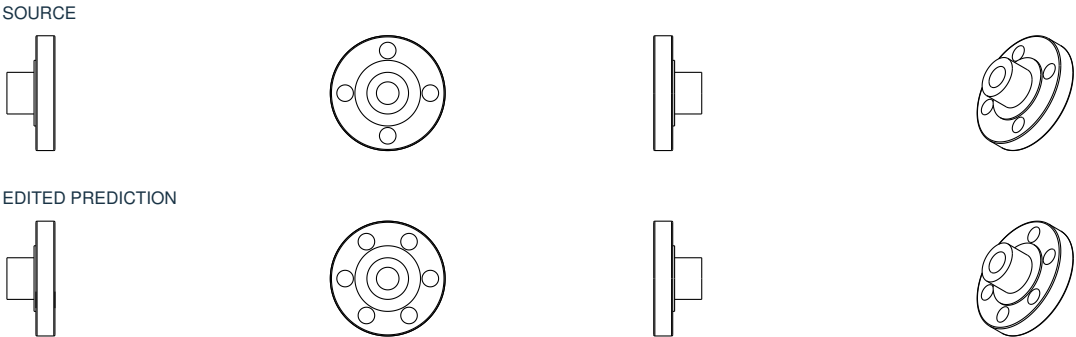
Assumed $E = 210,000 \text{ MPa}$, $\nu = 0.3$; static 1 N test. Stress convergence / operating safety are not established. Surface edges display the tetrahedral boundary, not every interior element.

Figure 7. **Large FEM method walkthrough with actual mechanical edit examples.** The final 3B mounting-angle and six-hole flange predictions supply all meshes and fields. The left column displays the tetrahedral boundary mesh; the middle shows displacement magnitude; the right shows diagnostic von Mises stress, clipped at the per-example 95th percentile. Blue nodes are fixed and red arrows show the resultant load direction. Independent CAD/reference agreement, equilibrium, energy and mesh checks accompany these plots; stress color alone establishes neither correct editing nor strength. The listed values use each saved fine mesh and the disclosed synthetic static scenario.

Actual CAD-derived drawings: source and final 3B edited prediction



Flange: 4 to 6 mounting holes; same diameter and bolt-circle radius



Projected views are autoscaled; dimensions and response must be read from CAD / solver artifacts.

Figure 8. **Linked engineering drawings from actual mechanical CAD solids.** Front, top, right and isometric views show the original and final 3B predicted mounting angle and flange. The flange edit changes four mounting holes to six equally spaced holes while retaining their diameter and bolt-circle radius; each panel is independently autoscaled. Both edited predictions pass the disclosed static geometry/FEM diagnostics. The views contain no inferred stress, production dimensions or hidden geometry repair.

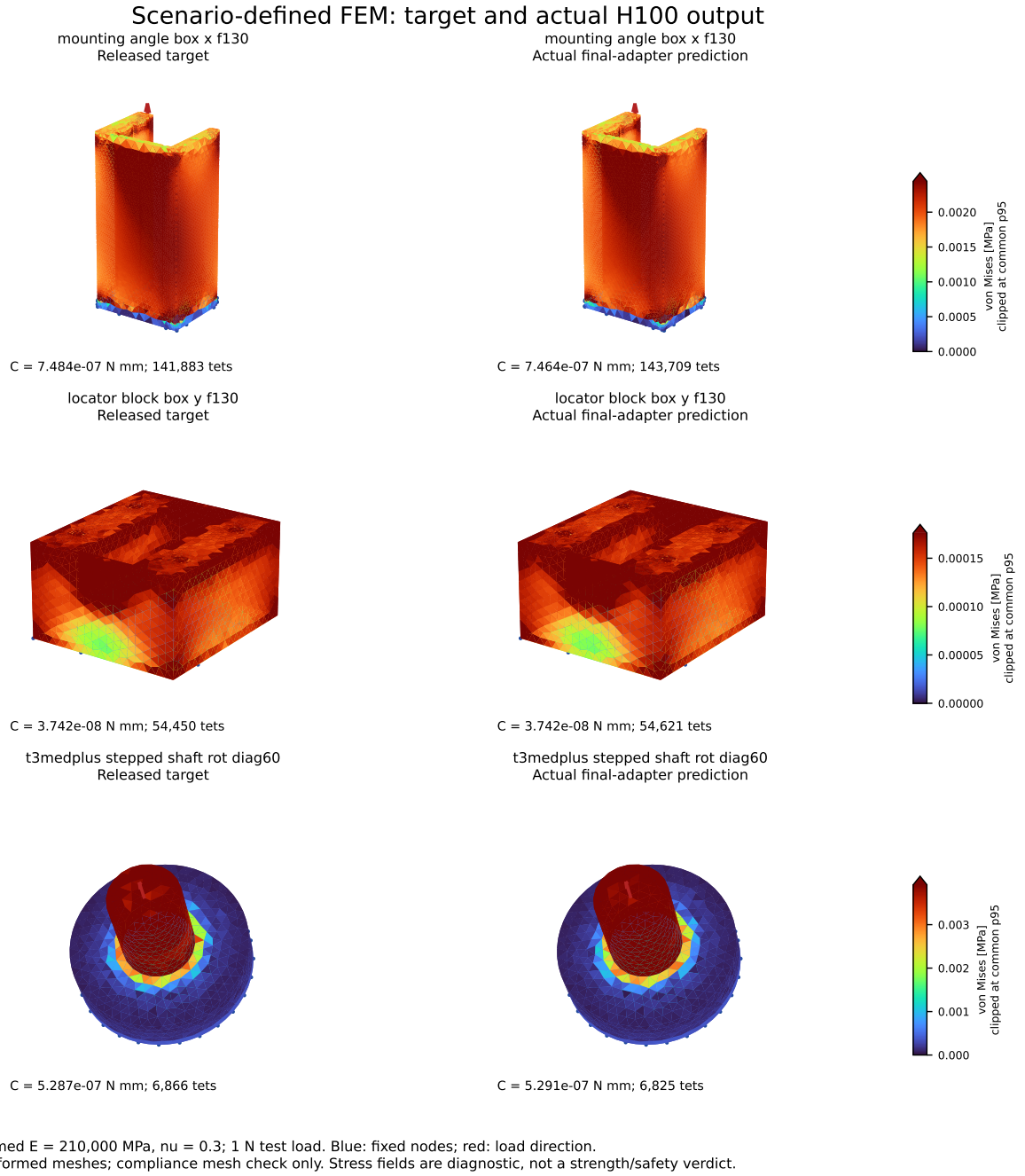


Figure 9. **Actual FEM fields for three geometry-matching H100 edits.** Released targets and model outputs use the documented synthetic test contracts. Blue points identify fixed nodes; arrows indicate the load direction. Undeformed surface stresses share a scale within each row, clipped above the common 95th-percentile value. Compliance passes the mesh-sensitivity check; stress colors are diagnostic rather than a strength verdict.

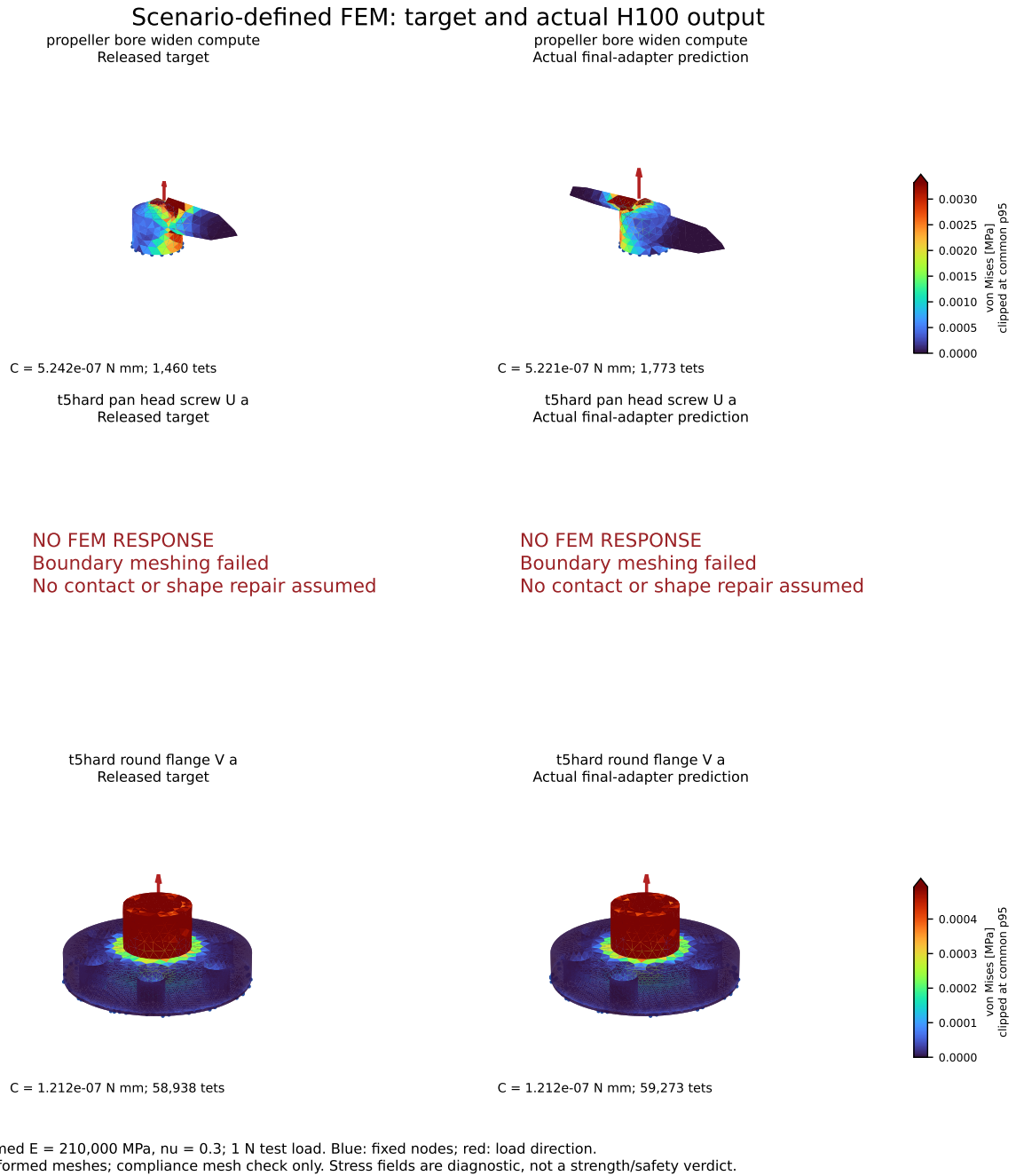


Figure 10. **FEM diagnostics do not replace edit fidelity.** Incorrect propeller and flange shapes produce similar compliance under the chosen test load, but fail strict reference geometry. The screw row has no FEM field because meshing fails; blank panels are explicitly marked. The same stress-scale and fixture conventions apply as in Fig. 9.

E. Focused visual audit details

The fixed case list comprises mounting angle, locator block, shaft, propeller, screw and round flange, previously chosen for illustration. Actual saved STEP files are tessellated, hashed and rasterized with the common-camera contract in Section 3. The earlier H100 adapter supplies six available solids; the final 3B adapter supplies four completed outputs. The adapters differ in model size, training corpus and optimization, so this subset is not a controlled model comparison. Failed/incomplete final outputs remain unavailable. The mean absolute per-case change in edit-mask IoU be-

Saved outputs	Available	Silhouette IoU	Edit-mask IoU
Earlier H100	6/6	0.911828	0.584674
Final Colab 3B	4/6	0.999993	0.874967

Table 7. **Diagnostic projected means at 512 pixels.** Average views within cases, then cases with available outputs. Undefined empty-union edit views are excluded; unintended nonempty changes remain. Unequal availability and raster-degenerate masks prohibit reading this table as general comparative accuracy.

tween 512 and 1,024 pixels is 0.000113 for the older outputs and 0.000033 for the final available outputs. This small numerical change does not prove metric validity. For example, the flange’s reference isometric change mask has only one pixel at 512 resolution. That degenerate view lowers the final flange’s edit mean to roughly 0.5 despite essentially exact solid agreement, and contributes to the final aggregate of 0.874967. No mask-area threshold is introduced after inspecting results. Future full-test protocols should define minimum visible-change area or feature-aligned views in advance. Occluded features require solid/feature measurements; silhouettes cannot replace them. The raw report records every view’s reference-change pixel count, two resolutions, all source/target/prediction STEP hashes and unavailable outcomes.

F. Earlier single-source dataset protocol

Published pairs. We use the `edit-bench` configuration of `BenchCAD/BenchCAD`, pinned at revision `5919f578ab09ec283603a082fab07c7639ab56eb`. It contains 748 instruction/source/target pairs with original and target STEP data [16]. The dataset declares CC-BY-4.0. Our downloaded parquet checksum and source-record identifiers are retained. Pipe, duct, fitting, manifold, valve, hose, resistor, circuit and PCB family names are excluded, removing 38 records and leaving 710 mechanical edit pairs.

Isolation before training. A complete mechanical family is one initial group. Families whose source or target ASTs share a template after masking numeric literals are joined into a component. Hash-ranked components are assigned

approximately 70/10/20 to training, validation and test before normalization, patches or augmentation. This gives 533/50/127 pairs. Family and component identities are disjoint across splits; no target from a held-out family enters fine-tuning. Template isolation is conservative but does not prove all cross-dataset geometric ancestry is disjoint. The fresh run does not load the earlier mechanical adapter.

Context-limited coverage. We retain a pair only when its full chat, including target, fits 3,072 tokens and its target patch fits 512 tokens. The rule is applied once before baseline evaluation and training. It retains 492/44/126 train/validation/test pairs, excluding 48 records across all splits. No example is silently truncated. The one excluded test pair and all excluded training examples remain identified in the run manifest. Results apply to the retained slice rather than the entire release.

Protocol change. The author labels this edit subset a held-out benchmark. We deliberately re-partition it into local training and evaluation data to satisfy the present study’s training objective. This invalidates claims of official benchmark-test generalization or direct leaderboard comparability. The local held-out components remain unused for optimization and checkpoint selection.

Reference-quality limits. Readback finds one invalid rivet STEP among the 126 retained test references. We also identify a concrete instruction/target conflict: `topup_stepped_shaft_top_slot_rev` requests cutting a blind slot, but its source already has the slot and its target removes the operation. Published labels are retained without silently repairing them; this case and its source diff are recorded. The experiment measures agreement with released targets, not certified compliance with every natural-language request. We do not infer that every reverse-edit record has this defect.

Earlier drawing pilot. A separate sample of 2,000 `BenchCAD` code-generation parts yielded 150 verified local numeric-edit pairs from 76 hard parts and 40 families, partitioned 108/22/20 by family. It requires at least eight CAD calls and compound geometry, and excludes external edit-program templates after masking literals. Chamfer, hole, extrusion and fillet edits contribute 62, 48, 24 and 16 pairs. The sample exports 904 referenced SVG views and 150 before/after sheets. Its instructions disclose the edit line and numeric answer; it is a transport feasibility pilot, not the main online benchmark.

G. Earlier 1.5B experiments and transport controls

Training. We start from a fresh 4-bit Qwen2.5-Coder-1.5B-Instruct conversion [4] and train a QLoRA adapter [3] using MLX on an Apple M5 with 16GB memory. The model conversion revision is b3252a2f97102b1fb1571fec2c9b27219a8536be. Seed 17, batch size one, learning rate 10^{-4} and the last eight transformer blocks are fixed. Adapters have rank 8, scale 20, no dropout, and cover attention and MLP linear projections in those blocks. Prompt tokens are masked from the loss. We run 492 optimizer steps, corresponding to one pass of the retained training rows, and use the final checkpoint. Validation is logged, not used to select among test-informed alternatives. Saved adapters are reloaded before evaluation.

Evaluation and metrics. Base and trained models receive the same original programs, published instructions and patch contract. Greedy decoding is capped at 512 new tokens. All 126 retained test records are evaluated in the same fixed order. Applicability requires a valid patch and syntactically valid program; AST equality requires the executed program to match the published target’s normalized syntax. Solid scoring executes the actual prediction and compares it with the released STEP reference. No gold edits repair a model prediction.

System	Patch	AST	Broad	Strict
Base, strict interface	0	0	0	0
Base, bare-list transport control	56	0	1	0
Fresh online-pair QLoRA	77	12	18	12

Table 8. **Local online-pair benchmark: counts out of 126.** All systems use the same family/template holdout. The transport control only wraps a bare JSON edit list into the required object, without changing coordinates or content. Timeouts and comparison errors remain separately disclosed; they are not measured incorrect geometry. This is a local split, not an official BenchCAD score.

Transport control. The base model frequently emits a bare list of edits rather than the required object containing edits. We therefore add a deterministic diagnostic that wraps a syntactically valid bare list while preserving every operation. It makes 56 of 126 base predictions applicable, but none matches the target AST. This control distinguishes one schema mismatch from the remaining content and coordinate errors; it does not rewrite geometry, infer missing operations or correct indexing.

Results and operational coverage. The trained model produces 77/126 applicable patches and 12/126 exact tar-

Category	Cases	Target AST	Strict match
T1	31	11	11
T2	18	1	1
T3	18	0	0
T4	34	0	0
T5	25	0	0

Table 9. **Trained results by published edit category.** Category IDs are retained from the source; counts are not pooled across different evaluation contracts.

get ASTs. Of 125 independently valid STEP references, 39 predictions build valid solids, 18 reach $\text{IoU} \geq 0.95$, and 12 reach strict $\text{IoU} \geq 0.99999$ (9.6%). The bare-list base control builds 3 solids, with 1 broad and 0 strict matches. Thus the experiment exposes a substantial remaining gap: most held-out edits fail to produce the reference geometry, particularly beyond literal-parameter changes. Independent reference readback finds 125/126 valid target solids. One released rivet reference is invalid and is excluded from the geometric success denominator, while all task outputs remain reported. For the trained run, 39 cases are geometrically scored, 37 have execution errors and 49 have invalid patches; there are 0 timeouts and 0 Boolean-comparison errors. The observed code failures include invalid feature operations and damaged source coordinates; they are distinct from valid solids with low IoU. We keep base and trained outputs, patch errors, executed-solid measurements, reference hashes and category breakdowns. One seed and one model size cannot establish a robust scaling law or broad superiority over CAD agents. No official benchmark claim or statistical population failure rate is inferred from this local slice.

Additional H100 run. After the local experiment, available server compute supports a separately reported CUDA run. A fresh unquantized BF16 Qwen2.5-Coder-1.5B-Instruct model is trained on the identical 492 retained training rows for three epochs (1,476 updates) on one NVIDIA H100 NVL. PEFT adapters use rank 8, alpha 160 (scale 20), no dropout and the same seven attention/MLP projections in the last eight blocks. Seed, batch size, learning rate, prompt masking, context limit and greedy generation limit remain unchanged. We reload the final adapter and evaluate all 126 cases; no test-based checkpoint selection is performed. Different precision, software implementations and epoch counts prevent interpreting the comparison as an isolated hardware effect or an independent preregistered confirmation.

The final CUDA adapter yields 61/126 applicable patches and 16/126 target AST matches. Against 125 valid released STEP solids, 36 predictions execute, 19 meet broad IoU and 15 meet strict IoU (12.0%). There are 65 invalid patches, 24 execution errors, 0 timeouts and 0 comparison

System	Patch	AST	Broad	Strict
CUDA base	0	0	0	0
CUDA base, list wrapper	47	0	0	0
CUDA final LoRA	61	16	19	15

Table 10. **Additional three-epoch H100 experiment.** Counts cover the same 126 retained tasks and 125 valid references, using the unchanged local geometry thresholds. Both base and final model are decoded with the CUDA implementation; the wrapper control preserves predicted edit contents.

errors; the same invalid released reference is excluded from the geometry denominator.

Localized drawing pilot. The earlier 100-step adapter retained 92 training, 22 validation and 12 test examples after context filtering. All 12 held-out exact patches execute and match their references. Under a unique-substring line parser the base model already matches 11/12 solids. That result mainly improves transport formatting and cannot support a claim of advanced CAD reasoning. The online experiment uses published instructions without line hints, a fresh adapter and its own held-out families.

Completed multi-source Colab extension. A separate fresh 3B QLoRA adapter trains on 2,540 paired edits from BenchCAD and CAD-Editor. On 252 held-out tasks it yields 173 applicable patches, 139 executed solids and 39 strict geometry matches (15.48 %), versus one strict match for its fresh base. Appendix D reports the full protocol, source-specific results and new-adapter FEM audit. This extension uses a different model and corpus; its counts are not pooled with Table 10.

H. Separate structural and clearance controls

The full CAD benchmark above checks mechanical shape. A separate six-case FEM audit is described in the supplement. We retain a separately identified deterministic SVG route to show what physical or manufacturing checks require when a drawing does supply a contract. These synthetic controls are not training data or accuracy evidence for the learned BenchCAD model.

Axial trusses. An importer reads supported untagged truss geometry, section dimensions, material notes, nodal loads and supports. A bounded request interpreter generates validated SVG tree patches. Re-importing the actual patched SVG determines member lengths, orientations and axial areas. Linear-elastic pin-jointed elements use EA/L stiffness; assembled displacements, stresses and reactions are recomputed. The numerical verdict checks equilibrium under small-displacement and nodal-load assumptions, not buckling, fatigue, joints or material strength.

Route	Edits	Outcome
Axial trusses	20,008	Equilibrium passes
Perforated plates	19,992	699 clearance violations

Table 11. **Historical structural pipeline audit.** All 40,000 supported edits match their stored target trees. These counts describe a deterministic synthetic pipeline, not a learned CAD benchmark. The plate violations are faithful edits flagged by the geometric checker.

Plate clearances. A separate importer recovers a rectangular boundary and circular-hole centres, diameters, thickness and stated limits. It computes overlap, edge distance and ligament. A faithful hole-enlargement request can reduce clearance below the printed minimum. A geometric violation is an engineering consequence of a correctly executed request, not necessarily an edit error; no continuum plate stress follows from a clearance score.

For example, increasing all truss member depths from 20 to 40 mm changes the recovered area and reduces peak axial stress from 11.861 to 5.930 MPa under the same load. Enlarging plate holes from 10 to 16 mm at fixed centres 20 mm from a boundary reduces edge clearance from 15 to 12 mm and fails a 15 mm rule. These are distinct checks with explicitly stated assumptions. The saved audit and replayed source/edited SVGs remain separate from the new online training run.

When FEM is applicable. A stress, stiffness, buckling, thermal or contact request needs physical units, constitutive properties, loads and boundary conditions, followed by solver and mesh checks. The inspected BenchCAD edit pairs do not provide this analysis contract. We retain *not evaluated* for the full released benchmark and distinguish the added six-case, synthetic-load FEM audit from validation under actual operating conditions.

Verification coverage. All six illustrated H100 cases are actual learned CAD program edits, followed by execution and comparison with released STEP targets. Each is now submitted to a separate 3D FEM response and mesh audit under the documented synthetic contract. The online benchmark and localized mechanical pilot check geometry; the plate controls check clearances. The six-case supplement solves 3D linear-elastic FEM problems, while the separate truss controls solve axial FEM equilibrium. Geometry agreement does not establish stress, stiffness or safety. FEM for a mechanical part would require an explicitly specified material, unit system, load case, supports and a validated mesh/solver setup, which are absent from these released edit pairs.